Balancing FAIR Data Principles with GDPR Compliance: A Guide for Researchers

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Data Lifecycle model

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CESSDA Training Team (2017 - 2022). *CESSDA Data Management Expert Guide.* Bergen, Norway: CESSDA ERIC. Retrieved from https://dmeg.cessda.eu/

Principles and / or legal obligations

The FAIR principles are not directly part of the GDPR, but they are often used in conjunction with it, particularly when dealing with research data and health data.

The **GDPR (General Data Protection Regulation)** and the **FAIR data principles** both relate to data governance, but they come from different domains with distinct goals:

- **GDPR** is a legal framework aimed at protecting personal data and privacy of individuals in the EU.

- **FAIR principles** are a set of guidelines to ensure data is **Findable, Accessible, Interoperable, and Reusable**—mainly within scientific and research contexts.

FAIR principles

An article published in Nature in 2016*, describing the basic principles which should be followed for sharing research data in sciences of all kinds.

Findable: the first step in (re)using data is to find them. Metadata and data should be easy to find for both humans and computers. Machine-readable metadata are essential for automatic discovery of datasets and services.

Accessible: Once users find the required data, they need to know how they can be accessed, possibly including authentication and authorisation.

Interoperable: The data usually need to be integrated with other data. In addition, the data need to interoperate with applications or workflows for analysis, storage, and processing.

Re-usable: The ultimate goal of FAIR is to optimise the reuse of data. To achieve this, metadata and data should be well-described so that they can be replicated and/or combined in different settings.

https://www.go-fair.org/fair-principles/

* Wilkinson MD, Dumontier M, Aalbersberg IJJ, Appleton G, Axton M, Baak A et al. The FAIR Guiding Principles for scientific data management and stewardship. Scientific Data. 2016;3. 160018. https://doi.org/10.1038/sdata.2016.18

Source

GDPR principles

- **Lawfulness, fairness, transparency**: Be open about what you are doing, and make sure that you follow applicable national laws and regulations
- **Purpose limitation**: Define a specific purpose for the processing of personal data. The GDPR acknowledges it is not always possible to fully identify the purpose at the time of data collection
- **Data minimisation**: Only collect as much personal data as needed for the purpose
- **Accuracy**: Personal data should be accurate
- **Storage limitation**: Only keep personal data as long as necessary for the purpose
- **Integrity and confidentiality**: Ensure that the data is safe

Source: Art. 5 of the GPDR: https://gdpr-info.eu/art-5-gdpr/

Linking

Aspect	GDPR	FAIR Principles	Link
Purpose	Protect individuals' personal data	Enable efficient data reuse and sharing	Both promote responsible data stewardship
Data Scope	Personal data (e.g., names, emails, identifiers)	Any data, including non-personal scientific data	FAIR may include personal data, so must comply with GDPR
Data Accessibility	Only with lawful basis and privacy safeguards	As open and accessible as possible	FAIR must balance accessibility with GDPR's restrictions
Data Reuse	Requires purpose limitation and data minimization	Encourages broad reuse with rich metadata	GDPR-compliant reuse must include legal/ethical considerations
Metadata & Provenance	Encouraged for accountability and consent tracking	Central to being Findable and Reusable	Shared emphasis on data traceability and transparency

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Synergies – Where GDPR and FAIR Align

Area	Explanation
Metadata & Provenance	FAIR requires detailed metadata for data discovery and reuse. GDPR encourages metadata to document consent, data origin, and processing purposes. Both promote traceability and accountability.
Transparency	FAIR promotes open, well-documented data. GDPR requires transparency about how personal data is collected and used. Shared commitment to clarity.
Data Stewardship	FAIR principles push for sustainable data management. GDPR emphasizes responsible handling of personal data. Both frameworks drive the professionalization of data management.
Machine-Readability	FAIR encourages machine-readable formats for interoperability and reuse. GDPR supports machine-readable formats (e.g. for data portability).
Reusability with Conditions	FAIR allows reuse “as open as possible, as closed as necessary.” GDPR mandates limitations for personal data. Both allow reuse within appropriate constraints.

Tensions – Where GDPR and FAIR Conflict

Area	Explanation
Open Access vs. Privacy	FAIR encourages open access to data. GDPR restricts access to personal data without legal basis and consent. This is the biggest point of conflict.
Findability vs. Anonymity	Making data easily findable can risk re-identification, especially with rich metadata. GDPR requires minimizing risks of personal data exposure.
Metadata Sensitivity	Metadata in FAIR may inadvertently reveal personal data or identifiers (e.g., through dataset descriptions or provenance records).
Long-term Storage vs. Storage limitation	FAIR supports long-term data retention for future reuse. GDPR requires not storing personal data longer than necessary.

Resolving the Tensions: A step-by-step guide

1. **Know your data:** Assess the degree of sensitivity and whether (parts of) the data are regulated by national law. This can be done in a Data Protection Impact Assessment (DPIA). Consult your DPO.

2. **Can data easily be anonymized, pseudonymized, or will this be impossible?** Minimize the amount and sensitivity of personal data published, while maintaining a dataset that actually contains useful information.

3. **Think FAIR from the start!** Check where the data could be archived/published and what the requirements are to ensure F, A, I and R. Include data use licenses and access controls in metadata.

4. **Draft your participant information letter:** Make sure that descriptions of purpose, duration of processing, contact information, future data access and re-use cover the entire lifecycle of the data. Consult your research support.

5. **Use controlled vocabularies and standards** that integrate both FAIR and legal compliance requirements (e.g., GDPR metadata concepts, DDI metadata standard).

6. **Know where your data is, and keep it safe!** Build in data governance processes to evaluate FAIR compliance in light of legal and ethical obligation. Use Data Management Plans and agreements such as Data Processor Agreements to ensure control over the data .

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Resolving the Tensions: Practical Strategies

Case I:

Large survey on political opinions, large amount of demographic and background variables, special categories of personal data, adult participants

Know your data	Relatively high degree of sensitivity. Unintended exposure of personal data would involve a high risk for the rights and freedoms of the participants. Discuss your plans for publication and re-use early in the process with your research support and/or Data Protection Officer. Assess risks, if necessary in a DPIA.
Anonymisation	Anonymisation techniques can be applied without compromising data quality too much. Minimal risk of re-identification after anonymization
FAIR	Make the data findable and interoperable: Strike a balance between rich metadata and anonymity. Make the data accessible and re-usable: Clear licensing and rules on access controls.
Information	Inform your participants about the publication, but let them know that the final dataset will be anonymised and no longer subject to GDPR (e.g., deletion no longer possible)
Know where the data is	Clear and detailed Data Processor Agreements with survey agencies. Ensure secure data storage

Resolving the Tensions: Practical Strategies

Case II:

Qualitative research, interviews with public officials. Topic is not particularly sensitive, but anonymization is difficult.

Know your data	Relatively low degree of sensitivity.
Anonymisation	Anonymization difficult due to the topic and position of the respondents. Anonymization would significantly reduce the re-use value of the material. Can pseudonymization reduce some of the risk?
FAIR	Make the data findable and interoperable: Strike a balance between rich metadata and anonymity. Make the data accessible and re-usable: Clear licensing. Access controls particularly important for identifying material. Track and document re-use.
Information	Good information about the possibility to be identified in the material. Gather consent for publication!
Know where the data is	The data is subject to GDPR during its entire lifecycle. Facilitate and monitor requests by data subjects. Ensure integrity and confidentiality.

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Thank you!

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